



Matteo Martinelli

Management Engineer

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Available to work abroad

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Professional Experience

- **Nov 2024 – Now: Freelance professional trainer**

Teaching topics: Programming logics and design tools, Python, Java and OOP programming, data analysis with Python, Docker and containerization, principles of cloud computing, Artificial Intelligence and Machine Learning principles, Git and GitHub usage.

- Programs:
- IFTS Software programming with AI tools - 2026 edition, organized by IFOA
 - MECFUTURE-TRANSITION_PRO - 2025 edition: "Information Systems Innovation: IoT, Machine Learning, and Artificial Intelligence", organized by W-Training
 - Full-Stack-Dev – Application Design and Development Technician - 2025 edition, organized by IFOA;
 - ITS DevSecOps - 2024/2025 and 2025/2026 editions, organized by Fondazione Olivetti and promoted by IFOA;
 - Women and Digital – 2024 edition, organized by IFOA;
 - ICT Skills for Employment, Inclusion and Sustainability – 2024 edition, organized by IFOA.

- **Nov 2024 – Now: Postdoctoral Research Associate – Research group of Pervasive Computing, UNIMORE**

Research topics:

- Digital Twins, basic concepts, state of the art, gaps identification, open challenges contribution;
- Physical and digital interoperability, heterogeneity management, hierarchical architectures, Cyber-Physical Systems augmentation, twins orchestration, AI relationships;
- Digital Twins and industrial applications, patterns of specialization, use cases investigation, AI pipelines and applications, synergies with constrained industrial devices, Digital Twin design techniques for industrial systems.

Related activities

- Research group activities **dissemination** in the **research and business domain**;
- Industrial projects designs support for industrial research projects;
- Coordination of thesis students in research and experimentation activities;
- Organization of the IDETwin 2026 Workshop in Las Vegas, dedicated to advancing research and innovation in Industrial Digital Twin and Cyber-Physical Systems Intelligence.

- **Sept – Dec 2024: holder of the course "Data Science and Management", contract professor at the University of Modena and Reggio Emilia, Italy**

Teaching topics:

- Python and OOP recall, data manipulation, preparation and visualization for ML with Numpy, Pandas, and Matplotlib;
- Big data concepts, from traditional to NoSQL DBs, distributed file systems principles and computation, ERP systems;
- Provision of evaluation examinations.

- **Dec 2020 – Nov 2024: Research Grant** – Research group of **Pervasive Computing**, **UNIMORE**
 Research topics:
 - eXplainable AI (XAI) state of the art, causality learning principles, autonomous development of smart agents;
 - Application of XAI in industrial contexts, preliminary studies of causality for industrial root cause analysis, testing and simulations;
 - Study of the role of Digital Twins for industrial contexts, synergies between AI and Digital Twins, patterns of interactions in Cyber Physical Production Systems (CPPS).
 Related activities
 - Research group activities dissemination in the research community at Italian and international scientific conferences;
 - Contract teaching activities and teaching support.
 - **Nov 2019 – Dec 2020: Focus Improvement & Safety Specialist**, Full time – Ognibene Power
 Main activities: **Micro-areas management** and KPIs monitoring, Kaizen management, **SMED**, **Workplace Organization Step 1–3**, **Operation Tag**, **EWO**, **suggestions monitoring** and management, project management, **S-EWO**, **Green Tags**, Unsafe Conditions/Acts, S-Matrix monitoring, Green Cross monitoring.
 - **Jun – Nov 2019: Workplace Organization Support**, Part time – Ognibene Power
 Main activities: **Operation Tag analysis**, assembly layout analysis and improvement, tooling card analysis and improvement, **time and methods analysis**.
 - **Jan – May 2019: Master's Degree Internship**, Full time – Dana Motion System
 Main activities: **Assembly line analysis**, As Is State and Performance mapping, **methods and time re-engineering**, **Plan For Every Part**, **supplier coordination** and solution implementation, **Poka-Yoke POC** engineering based on **Convolutional Artificial Deep Neural Network**.
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Education & Training

Dec 2020 – May 2024: Ph.D. in Industrial Innovation Engineering – Research group of **Pervasive Computing**, **UNIMORE**

Main topics: **AI** applied to **IoT** scenarios, **Digital Twins** for **Industry** and **Manufacturing**, application of intelligence in industrial contexts.

Thesis: Smart Digital Twins in Industry 4.0.

Oct 2022 – Apr 2023: Guest Researcher at ISW - *Institut für Steuerungstechnik der Werkzeugmaschinen* (Institute for Technologies of machine tools control) 

Topics: **Digital Twins for Industry 4.0** and *Composability* - Universität Stuttgart, Stuttgart, Germany.

Jun 2022: ACAI TAILOR Summer School 2022 – Prize for the best PhD project presentation 

Topics: **Explainable and Trustworthy AI** - Universitat Politècnica de Catalunya, Barcelona, Spain.

Oct 2019: Management Engineering Master's Degree, **UNIMORE** 105/110

Thesis: *"An assembly line re-engineering: The Dana Motion System case"*.

Technologies applied: Lean Manufacturing principles, Spaghetti charts and layout analysis, operation activities classification (VA, NVA, SVA, Yamazumi), solution design, First Time Quality analysis, data collection and dataset design, Convolutional Neural Network training and test, Intelligent Poka Yoke setup.

Jul 2017: participation at the Erasmus+ Program – Krakow, Poland 

Dec 2015: Management Engineering Bachelor's Degree, **UNIMORE** 95/110

Thesis: *"Implementation of a reading and translation system of DXF files into two axis PLC instructions"*.

Technologies applied: PLC, cartesian robot, C#, code design, design patterns.

Computer Skills:

- Programming Languages: C#, **Java**, **Python**, **JavaScript**.
- Web technologies: **HTML**, **CSS**.
- Programming Libraries: Keras, Numpy, Pandas, Scikit-Learn, PuLP.
- DB Technologies and developing tools: SQL, document-oriented DBs, Docker, Git, UML.
- Operating Systems: Windows, Ubuntu, MacOS.
- Software: Anylogic – Material Handling and Process Model libraries, Ms Office suite.

Languages

English (C1 grade);

German (A2 grade);

Italian (mother language).

Soft Skills

- Precise
- Reliable
- Communicative
- Loyal
- Open minded, *learning* and *people* oriented
- Assertive

Other Information

- Attachment 1: Certificates and Badges.pdf

Date

06/01/2026

Matteo Martinelli



Certificates and Badges



Software and technologies for Data Science

- Date: March 2021;
- Provided by: the Big DataLab project sponsored by the ESF.

https://openbadges.bestr.it/public/assertions/rh6U8Z45S8GyXH_Wx6koSg



Internet of Things and Predictive Analysis

- Date: June 2021;
- Provided by: the Big DataLab project sponsored by the ESF.

<https://openbadges.bestr.it/public/assertions/JNZKBoomR8GTvl-0YMc3IQ>



Deep Learning Applications

- Date: June 2021;
- Provided by: the Big DataLab project sponsored by the ESF.

<https://openbadges.bestr.it/public/assertions/bVoUXhQHRG2vhfunvjMTCw>

Date

06/01/2026

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